

Total No. of Questions : 10]

SEAT No. :

P2350

[Total No. of Pages : 3

[5254]-683

**B.E. (Information Technology)**

**MACHINE LEARNING**

**(2012 Pattern) (End Semester)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Use of calculator is allowed.*
- 4) *Assume Suitable data if necessary*

**Q1) a)** For following multi-class classification predictions: **[6]**

| Predicted |    |    |    |
|-----------|----|----|----|
|           | 15 | 2  | 3  |
| Actual    | 7  | 15 | 8  |
|           | 2  | 3  | 45 |

Calculate Accuracy, Per Class Precision, Per Class Recall.

**b)** Define and explain regression with its model. **[4]**

OR

**Q2) a)** Derive and explain output code matrix for One Vs One and One Vs Rest Scheme for construction of Multi class classifier (for 3 classes) **[6]**

**b)** How the performance of a regression function is measured? **[4]**

**Q3) a)** What are the different types of regularizers? **[6]**

**b)** What is a Probabilistic Model? Give an example of it. **[4]**

**P.T.O.**

OR

**Q4) a)** What are grouping models and grading models? Give one example of each. [4]

b) Consider the following data points: [6]

| X | Y    |
|---|------|
| 1 | 1.5  |
| 2 | 2.75 |
| 3 | 4    |
| 4 | 4.5  |
| 5 | 5.5  |

Calculate the Cost Function for  $\theta_0 = 0.5$  and  $\theta_1 = 1$  using linear regression.

**Q5) a)** In a given dataset of 14 samples, 9 are positive and 5 are negative. Calculate the entropy of the dataset. [6]

b) Find all association rules in the following database with minimum Support =2, minimum confidence=70%. [12]

| TID | Items     |
|-----|-----------|
| 1   | {a,b,c}   |
| 2   | {b,c,d,e} |
| 3   | {c,d}     |
| 4   | {a,b,d}   |
| 5   | {a,b,c}   |

OR

**Q6) a)** Consider following dataset: [10]

| $X_1$ | $X_2$ | Y |
|-------|-------|---|
| 2     | 1     | 4 |
| 6     | 3     | 2 |
| 2     | 5     | 2 |
| 6     | 7     | 3 |
| 10    | 7     | 3 |
| 4     | 4     | 2 |
| 7     | 6     | 3 |

Model this function using the K-nearest neighbor regression. What will be the value of Y for the instance  $(X_1, X_2) = (4, 5)$  and  $K = 3$

b) How empirical probabilities can be used in ranking and probability estimation. trees? Explain the purpose of pruning the subtree of a decision tree. [8]

- Q7)** a) Explain one dimensional and N- Dimensional Gaussian Mixture Model.[8]  
b) What is logistic regression? How it outperforms basic linear classifier?[8]

OR

- Q8)** a) Write a note on Multi-nominal Distribution. [8]  
b) Define and describe Logistic Regression. [8]

- Q9)** a) Explain Random Forest Method. [8]  
b) What is the motivation behind Reinforcement Learning. Explain it with help of digram stating its important entities. [8]

OR

- Q10)** a) Write a note on Deep Learning and its applications. [7]  
b) Define and explain: [9]  
i) Sequence Prediction  
ii) Sequence Generation  
iii) Sequence Classification

